A DIGITAL MARKETPLACE FOR UPCYCLING AND VALORIZING SRI LANKA'S TEXTILE WASTE

INDIVIDUAL THESIS

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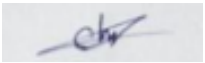
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ABSTRACT

Textile waste is a significant circular-economy challenge because material characteristics, physical condition, and potential recovery routes are difficult to assess consistently from manual inspection alone. Within the TexCycle AI digital marketplace, the Member 01 Fabric Intelligence component provides an image-based decision layer that transforms an uploaded textile image into structured information for downstream sustainability and marketplace services. The implemented component combines a 16-class fabric classifier based on EfficientNet-B3 with a separate defect-analysis pathway based on MobileNetV2. The fabric classifier accepts normalized 224×224 RGB images and predicts fabric type with a confidence score. The defect pathway predicts Defect or No Defect and provides an application-level defect signal that contributes to health assessment. Fabric-specific baseline lifetime information and the derived health score are then used to estimate repairability and remaining lifespan.

The development workflow covered dataset investigation, image preprocessing, class balancing, transfer learning, model training, validation, checkpoint selection, artifact management, FastAPI inference integration, seller-workflow integration, and functional testing. The latest documented evaluation reports 89.68% test accuracy for the 16-class fabric classification model and 90.87% accuracy for defect detection. A best validation accuracy of approximately 91.41% was also reported during the EfficientNet-B3 experiment before early termination. These values demonstrate technical feasibility on the available development/evaluation data; they do not establish universal performance on unseen industrial textile populations. The contribution is therefore an integrated and reproducible Fabric Intelligence pipeline for preliminary textile assessment, with clear interfaces to sustainability analysis and marketplace services.

Keywords: textile classification, computer vision, EfficientNet-B3, MobileNetV2, fabric defect detection

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I also thank the open-source machine-learning ecosystem, technical documentation, and scholarly publications that helped choose and use convolutional neural network techniques. The candidate is still in charge of the thesis's final execution, findings interpretation, citations, and assertions.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Computer Vision
F1	Harmonic mean of precision and recall
FastAPI	Python web framework for APIs
GPU	Graphics Processing Unit
ML	Machine Learning
RGB	Red, Green, Blue
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
UNEP	United Nations Environment Programme

1. INTRODUCTION

1.1 Overall project context

A digital marketplace called TexCycle AI was created to help Sri Lankan textile waste be upcycled and valued. In addition to providing AI-supported fabric comprehension, sustainability intelligence, commercial intelligence, and marketplace prediction, the expanded system links buyers and sellers of textiles. The separate parts are purposefully linked together: Material and condition characteristics are identified using Fabric Intelligence. Commercial Intelligence calculates reference value and appropriate buyer groups; Sustainability Intelligence transforms those attributes into environmental and circular indicators; and Marketplace Decisions about listing performance, sales time, and market trends are supported by predictive analytics.

The initiative is driven by the need to extend the useful life of textile materials and simplify the classification and routing of waste streams. Instead of considering abandoned textiles as residual trash right away, textile circularity literature stresses reuse, repair, recycling, and other measures that can maintain material value. [1, 2]. The first technical question is therefore useful in this system context: what can be deduced from a textile image before a seller moves on to the subsequent phases of the marketplace?

1.2 Individual component context

The Fabric Intelligence component is the responsibility of Member 01. It is not intended to take the place of an industrial inspection procedure or a textile laboratory. Rather, it offers a first computer-vision evaluation that may be produced instantly from an image submitted by the vendor. Fabric type, prediction confidence, fault presence, a health-oriented interpretation, repairability, and predicted remaining lifespan are the primary topics of this component. The subsequent Sustainability Intelligence process uses these results as structured inputs.

Table 1: Member 01 component responsibilities

Component	Individual responsibility	Primary output
Fabric classification	16-class image classification using EfficientNet-B3	Fabric type + confidence
Defect analysis	Binary defect/no-defect inference using the implemented defect pathway	Defect signal
Condition interpretation	Application-level interpretation of defect signal and image result	Health score
Repairability	Rule-based interpretation using health and fabric information	Repairability
Remaining lifespan	Fabric baseline × health-ratio style estimation	Estimated remaining life
Integration	FastAPI + seller workflow integration	Structured analysis response

1.3 Problem statement

When sellers submit diverse waste streams, manual identification of textile material and condition may be sluggish, unreliable, and challenging to scale. Before it can produce significant sustainability, price, or buyer-matching results, a marketplace that takes random textile postings also requires a minimal degree of material intelligence. Therefore, the construction of an image-based Fabric Intelligence pipeline that can categorize common textile categories and offer a preliminary condition signal in a format that the other system components can consume is the issue that Member 01 is trying to solve.

1.4 Research gap.

Fabric recognition and flaw detection are included in research on automated textile inspection; however, integrating these activities into a single seller-facing workflow presents a practical barrier for a circular economy. Surface imaging can facilitate automated fiber or fabric recognition, according to a textile recognition study. [3]. Convolutional models are also useful for detecting textile problems, according to defect-detection studies. [4, 5]. The operational integration of material categorization and preliminary condition intelligence with downstream circular economy services is the project-level gap that this component fills.

1.5 Aim and objectives.

Member 01's objective is to create, develop, and test an image-based Fabric Intelligence component for TexCycle AI marketplace preliminary textile categorization and condition evaluation.

- Create an EfficientNet-B3 transfer learning-based 16-class textile picture classification pipeline.
- Establish a distinct defect-analysis pipeline that may generate a Defect/No Defect outcome for textile photos that are uploaded.
- Using the provided project logic, translate model outputs into application-level health, reparability, and remaining-life data.
- Make the learned models available via an inference service that the application process may use.
- Use functional integration tests and suitable categorization criteria to assess the component.
- To prevent model outputs from being presented as laboratory-grade material certification, document restrictions.

1.6 Research questions.

- RQ1: To what extent can the textile categories included in the project dataset be classified by an EfficientNet-B3 model?
- RQ2: Is it possible for a lightweight defect-analysis model to offer a helpful initial condition signal for the seller workflow?
- RQ3: How can image-model outputs be converted into structured data that can be used by marketplace and sustainability components downstream?
- RQ4: What issues need to be resolved before the component can be verified against actual observations of industrial textile waste?

1.7 Scope, limitations, and assumptions

The TexCycle AI prototype's scope is restricted to image-based basic textile intelligence. The implementation makes no claims about identifying every potential fiber mix, form of contamination, chemical treatment, or microscopic fiber composition. The project's operational class vocabulary is the 16-class mapping.

Similarly, the stated defect result should not be construed as an industrial quality certificate because it is a model-level development result. Future directions, such as multi-image analysis, contaminant identification, color/pattern recognition, manual review, confidence levels, and active learning, are indicated in the project material.

1.8 Individual contribution

Dataset analysis and preprocessing, model-selection research, transfer-learning implementation, training and assessment, model artifact creation, inference integration, seller-workflow testing, result interpretation, and technical documentation are all included in the individual contribution. The contribution is different from the Sustainability Intelligence module, which is held by Member 02 and uses some of the Fabric Intelligence results.

1.9 Thesis structure

The component and research problem are introduced in Chapter 1. Chapter 2 examines pertinent systems and literature. The study and experimental techniques are presented in Chapter 3. Requirements, architecture, and design choices are covered in Chapter 4. Implementation, integration, and testing are covered in Chapter 5. The results and validation evidence are presented in Chapter 6. Interpretation, contribution, and restrictions are covered in Chapter 7. The work is concluded, and new directions are identified in Chapter 8. Appendices, glossaries, and references offer information about repeatability and supporting data.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope.

Four topics guided the scope of the literature review: transfer learning for computer vision, automated fabric-defect detection, image-based textile recognition, and textile waste and circularity. Peer-reviewed papers, reputable conference proceedings, and reliable circular economy sources were prioritized. Rather than asserting that the project replicates any one published system, the review was utilized to support technical decisions.

2.2 Textile waste and circularity

The goal of circular textile systems is to keep resources, components, and products at their maximum useful value for as long as feasible. According to the Ellen MacArthur Foundation, the shift to a circular textile economy necessitates modifications to the way goods are created, utilized, gathered, and processed. [1]. In the textile industry, UNEP also prioritizes enhanced value-chain processes, resource efficiency, and system-level circularity. [6]. Because material identification and condition evaluation can affect whether a textile is directed toward reuse, repair, recycling, or another recovery option, these concepts are pertinent to fabric intelligence.

The main conclusion for the market is that fabric type is not enough on its own. Depending on damage, contamination, reparability, and remaining usable life, a superficially comparable material may have a drastically different value. Therefore, rather than trying to tackle textile circularity through classification alone, the Fabric Intelligence component handles categorization and condition as complementary outputs.

2.3 Deep learning for image classification

Strong accuracy-efficiency trade-offs were shown by EfficientNet, which presented a compound scaling strategy that balances network depth, breadth, and input resolution [7]. For transfer-learning applications where a project requires a usable visual representation without building a very big network from scratch, this makes the

EfficientNet family appealing. EfficientNet-B3 serves as the foundation for the 16-class fabric classifier in the TexCycle implementation.

Transfer learning has a useful justification as well. Compared to the datasets required to pretrain contemporary convolutional networks, textile datasets available to a single project could be far smaller. While subsequent layers are tailored to the project-specific textile classes, lower-level visual aspects may be reused by starting with an ImageNet-pretrained representation.

2.4 Textile/fiber recognition

FabricNet examined many well-known CNN families and showed that surface imagery may allow the identification of a wide range of textile threads using convolutional architectures. [3]. The viability of collecting material-related information from fabric pictures is supported by such work, but it also draws attention to a crucial difference between marketplace image categorization and laboratory fiber detection. The lighting, size, backdrop, folding, occlusion, and material combination of a marketplace image may all change.

Thus, the TexCycle 16-class vocabulary is a categorization layer that is application-oriented. Blended, Chenille, Corduroy, Cotton, Crepe, Denim, Fleece, Leather, Linen, Nylon, Polyester, Satin, Silk, Terrycloth, Viscose, and Wool are the classes that have been documented for the implementation. Instead of demonstrating the precise fiber composition, the classifier should be seen as identifying the nearest supported project class.

2.5 Automated fabric defect detection

Because flaws impact product quality and might affect selections about downstream materials, fabric defect identification is a well-established computer vision issue. The survey literature highlights the ongoing significance of automated inspection and divides approaches into standard image-processing and learning-based methods. [4]. Lightweight architectures and cloud-edge deployment for real-time textile inspection have been investigated in more recent work. [5].

Using inverted residuals and linear bottlenecks, MobileNetV2 was created for effective computer vision under resource restrictions [8]. Because of its

effectiveness, it is a sensible design for a defect-analysis pipeline that eventually has to be served by an application. Instead of claiming full pixel-level defect segmentation in the application backend, the project's described implementation employs a binary Defect/No Defect interpretation.

2.6 Related techniques and comparison

Table 2: Comparison of related approaches

Approach	**Strength**	**Limitation for TexCycle**	**Project response**
Manual inspection	Human judgement and contextual knowledge	Slow and difficult to scale	Automated preliminary screening
Traditional image processing	Simple and interpretable for controlled settings	Sensitive to illumination and texture variation	Deep transfer learning
Large CNN trained from scratch	Task-specific representation	Requires more labelled data and compute	Transfer learning
EfficientNet family	Accuracy/efficiency trade-off	Still sensitive to dataset shift	EfficientNet-B3 + preprocessing
MobileNetV2	Lightweight inference architecture	May need task-specific tuning	Defect pathway
Laboratory fiber identification	High material specificity	Not equivalent to marketplace photos	Treat output as preliminary class prediction

2.7 Gap synthesis and conceptual framework

According to the review, there is a practical gap between standalone picture-recognition tests and a commercial mechanism that can convert visual evidence into structured data that can be used. Therefore, picture capture → preprocessing → fabric categorization → defect analysis → health interpretation → reparability and lifetime prediction → structured API response → downstream sustainability and marketplace services is the conceptual framework chosen for Member 01.

This framework is designed to be modular. While the defect model can be enhanced without restructuring the marketplace, the classifier can be retrained without altering the sustainability model contract. The division facilitates future replacement of individual inference components, repeatability, and model versioning.

3. RESEARCH METHODOLOGY

3.1 Research design.

An applied design-and-evaluation technique is used in the study. The component was created as a software artifact and assessed using functional integration tests and controlled model experiments. Problem conception, literature review, dataset inquiry, preprocessing, model selection, training, validation, model-artifact construction, API integration, application testing, and result interpretation make up the research cycle.

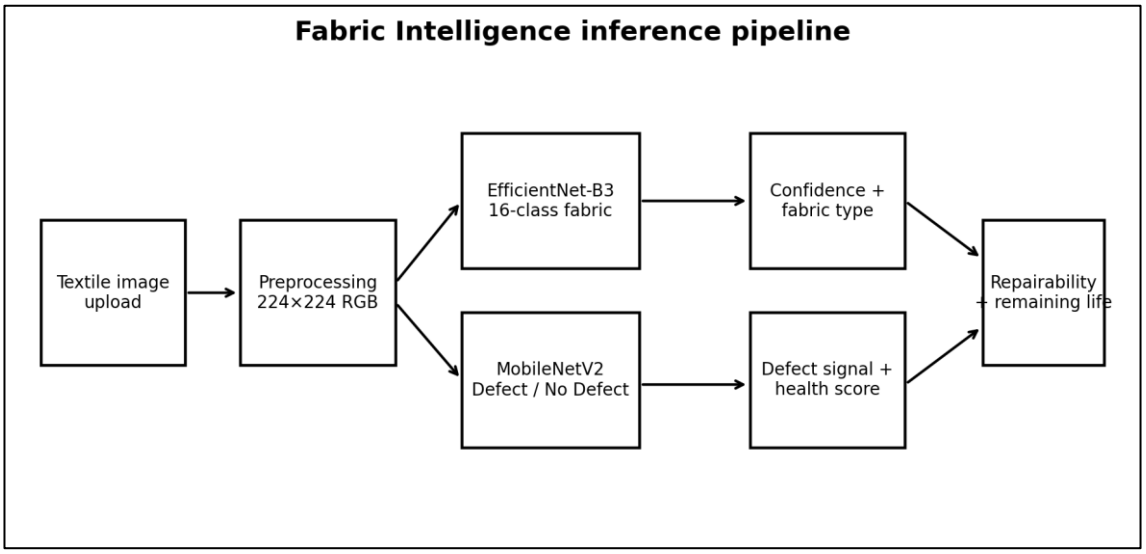


Figure 1: Fabric Intelligence inference pipeline.

3.2 Data and study materials

The project description identifies a 16-class fabric mapping and image-based training methodology. Rather than being presented as a representative census of Sri Lankan textile waste, the available development material is used as the project dataset. Class mappings, trained model checkpoints, and the inference service are examples of research artifacts.

Table 3: Evaluation metrics

Artefact	Role	Documented form
Fabric class mapping	Maps model output indices to project class names	fabric_classes.json

Fabric classifier	Stores trained 16-class model weights	fabric_classifier.pth
Defect model artefact	Supports defect-analysis pathway	Defect model artifact: repository documentation also references defect_unet.pth
Inference service	Serves model predictions to the application	Python / FastAPI
Frontend workflow	Presents analysis to the seller	React / Vite / TypeScript

The project description identifies a 16-class fabric mapping and image-based training methodology. Rather than being presented as a representative census of Sri Lankan textile waste, the available development material is used as the project dataset. Class mappings, trained model checkpoints, and the inference service are examples of research artifacts.

3.3 Experimental procedure

Data collection, cleaning, exploratory review, class filtering and balancing, image preprocessing, train/validation/test separation, baseline model, transfer-learning training, learning-rate scheduling, early stopping, best-checkpoint selection, evaluation, model saving, API integration, and application testing comprise the documented training workflow.

- A consistent RGB representation is created from the images.
- Pictures are scaled to the EfficientNet-B3 classifier's specified 224x224 input size.
- To prevent contaminating test or validation data, training preprocessing is maintained apart from evaluation preprocessing.
- When preparing and training datasets, class imbalance is taken into account.
- The detailed training process includes early stopping, learning-rate scheduling, and AdamW optimization.
- The checkpoint chosen for deployment is the one with the best validation performance.

- The inference service loads the chosen artifact and uses application requests to test it.

3.4 Measures and evaluation metrics

Accuracy is the primary reported parameter for multiclass fabric classification. Because a high overall accuracy might mask poor performance on specific classes, precision, recall, and F1-score are shown to be essential complementary criteria for publication-grade grading. Accuracy is the currently provided result for the binary defect route; for a thorough assessment, precision, recall, F1-score, and confusion-matrix analysis are advised.

Table 4 : Fabric Intelligence study artefacts

Metric	**Definition/purpose**	**Interpretation**
Accuracy	Correct predictions / all predictions	Overall correctness
Precision	Correct positive predictions / predicted positives	Controls false-positive interpretation
Recall	Correct positive predictions / actual positives	Controls missed-positive interpretation
F1-score	Harmonic mean of precision and recall	Balanced class-performance summary
Confusion matrix	Counts of predicted vs actual classes	Shows class-specific errors

3.5 Data analysis

Individual prediction confidence and model-level outcomes are kept apart. Validation accuracy is not the same as the confidence score displayed for a single uploaded image. As a result, the assessment solely considers single-image probabilities as inference evidence and gives model-level metrics. Error analysis should concentrate on visually related classes, mixed fabrics, folds, occlusion, shadows, odd backdrops, and minor faults.

3.6 Ethics, privacy, security, and responsible AI

Uploads of images may include identifying operational data or proprietary textile materials. Therefore, the program should limit access to authorized users, eliminate needless retention, and refrain from exploiting uploaded photographs for secondary uses without the proper authorization. The model's result should not be interpreted as an official laboratory assessment of the fiber's composition or quality; rather, it is advisory.

3.7 Reproducibility and research artefacts

Maintaining the class mapping, preprocessing rules, model architecture, trained checkpoint, training configuration, evaluation split, and inference preprocessing are all necessary for reproducibility. To link a marketplace listing to the model version that produced its analysis, model artifacts should be versioned. This keeps further model modifications from subtly altering historical interpretation.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 5: Functional and non-functional requirements

Requirement	Type	Description	Acceptance evidence
FR-01	Functional	Accept a textile image from the seller workflow	Valid image reaches inference service
FR-02	Functional	Return one of the supported fabric classes	Class label in response
FR-03	Functional	Return fabric confidence	Numeric confidence present
FR-04	Functional	Return defect result	Defect / No Defect response
FR-05	Functional	Provide health-oriented interpretation	Health value returned
FR-06	Functional	Estimate repairability	Repairability value returned
FR-07	Functional	Estimate remaining lifespan	Remaining-life value returned
NFR-01	Non-functional	Reasonable inference response time	Successful request under target environment
NFR-02	Non-functional	Deterministic preprocessing	Same image produces stable preprocessing
NFR-03	Non-functional	Model artifacts versioned	Artifact and mapping identifiable
NFR-04	Non-functional	Failure handling	Invalid inputs return controlled errors

4.2 Component architecture within the complete solution

The display layer, application backend, AI services, and data layer are all kept apart in the whole TexCycle design. Although it is called part of the selling workflow, Member 01 is located in the AI/ML service layer. Because of this division, the model service can be upgraded separately from the marketplace backend, and the

React/Vite interface can continue to be independent of model implementation specifics.

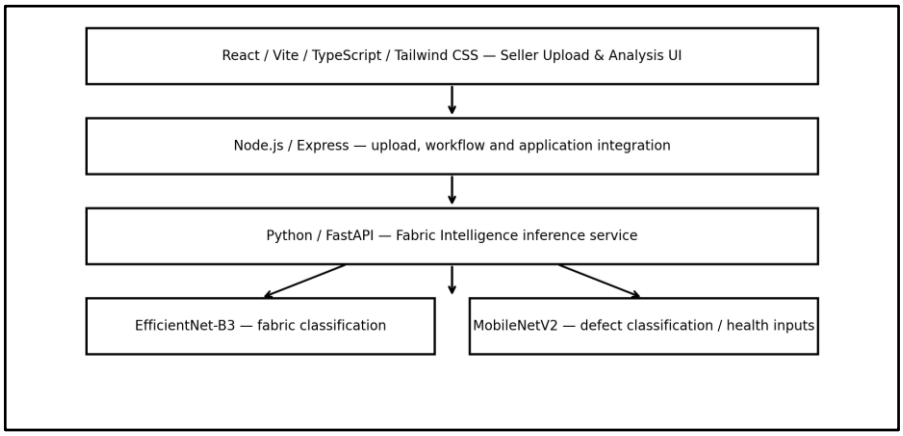


Figure 2 : Member 01 component architecture within TexCycle AI

4.3 Design alternatives and decisions.

Table 6 : Design alternatives and decisions

Decision	Alternative	Selected	Reason
Backbone	Custom CNN	EfficientNet-B3	Transfer learning and accuracy/efficiency balance
Defect model	Large detector	MobileNetV2 pathway	Lightweight deployment
Input	Variable-size images	224×224 RGB	Consistent model interface
Serving	Model inside Node backend	FastAPI ML service	Separation of ML and application logic
Output	Raw class only	Structured analysis response	Supports downstream modules
Condition	Pure manual entry	Model + rule interpretation	Reduces manual data-entry burden

4.4 Detailed design

4.4.1 Fabric classification

After receiving a normalized image tensor, the classifier generates a probability distribution across the 16 supported classes. The associated probability is kept as the confidence indicator, and the chosen class is the one with the highest probability under the trained model. With enough metadata for the seller interface and subsequent modules, the result is sent to the program.

4.4.2 Defect analysis.

A binary condition signal is provided by the faulty route. This signal serves as proof for health interpretation in the application-level design. The design purposefully refrains from asserting that accurate defect localization is comparable to a binary defect outcome. A segmentation model can be included behind the same service contract if pixel-level localization is eventually needed.

4.4.3 Health score

To determine a health-oriented score, the defined project logic analyzes fabric data and defect evidence. The health score is not a standard measurement used in textile engineering, but rather an application metric. Its goal is to give reparability and downstream sustainability analysis a consistent internal signal.

4.4.4 Reparability and remaining lifespan

The health outcome and fabric-specific baseline lifetime data are used to determine reparability and remaining lifespan. In theory, the recorded formulation is a baseline lifespan that has been modified by the estimated health ratio. The study uses these values as decision-support indicators, as they are calculated rather than explicitly measured.

4.4.5 Standards and professional practice

The design adheres to testable interfaces, clear API contracts, model versioning, separation of concerns, and modular software engineering concepts. With specified inputs and outputs, the picture models are handled like software components.

Engineering and research claims are kept apart, and acknowledged limits are recorded rather than concealed.

5. IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 7 : Development environment and technology stack

Layer	Technology/artefact	Purpose
Frontend	React, Vite, TypeScript, Tailwind CSS	Seller interface and result display
Backend	Node.js, Express.js	Application workflow and integration
AI service	Python, FastAPI	Model inference API
Vision model	EfficientNet-B3	16-class fabric classification
Defect model	MobileNetV2 pathway	Binary defect analysis
Image processing	Python imaging/tensor preprocessing	Resize, normalization, tensor conversion
Model artefacts	PyTorch checkpoints + class mapping	Deployment and reproducibility
Data layer	MongoDB / database-backed storage	Application records and analysis persistence

5.2 Component implementation

Training-time logic and inference-time logic are implemented separately. The training code records the evaluation results, optimizes the weights, builds the model, and prepares the dataset. After loading the trained artifact and class mapping and doing the same input preprocessing, the inference algorithm generates a structured answer. The possibility that deployment preparation will be different from evaluation preprocessing is decreased by this division.

An uploaded image can be accepted by the model-serving interface, which then provides structured analysis. The anticipated fabric type, confidence, defect outcome, and derived condition variables are included in a typical answer. The published API contract should be followed for the precise JSON property names; the thesis should not create endpoint fields that are not supported by the project specification.

5.3 Integration with the other components

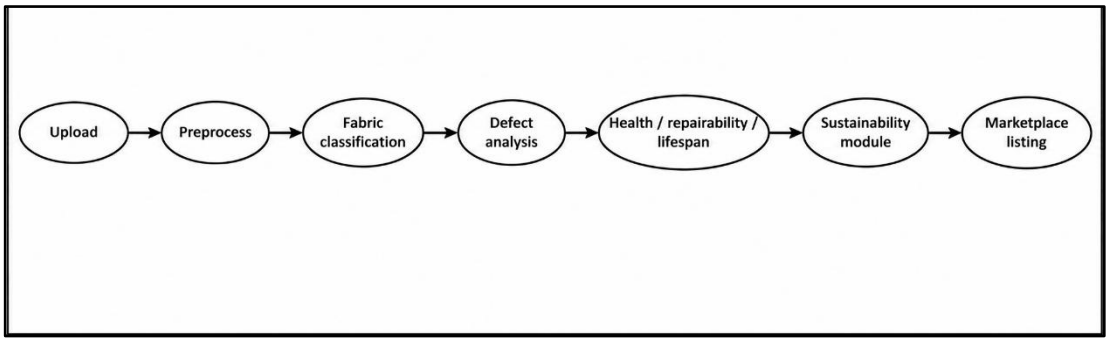


Figure 3 : Seller workflow and Member 01 integration

First, the vendor uploads a picture of a cloth. The signals for material and condition are generated by Fabric Intelligence. The fabric type, health score, reparability, and other relevant characteristics may then be consumed by the Sustainability Intelligence component. The generated textile data can be used in conjunction with marketplace and commercial intelligence services. By doing this, a sequential intelligence workflow is produced without requiring each AI component to process the raw image separately.

5.4 Verification and software testing

Table 8 : Verification and software testing

Test ID	Test objective	Expected result	Status basis
FI-T01	Valid textile image upload	Image accepted	Functional test
FI-T02	Unsupported file type	Controlled validation error	Functional test
FI-T03	Very small image	Preprocessing completes or controlled error	Robustness test
FI-T04	Fabric classification response	Class + confidence returned	Model integration
FI-T05	Defect response	Defect / No Defect returned	Model integration
FI-T06	Health derivation	Health value generated	Rule integration
FI-T07	Repairability derivation	Repairability generated	Rule integration

FI-T08	Remaining-life derivation	Estimated life generated	Rule integration
FI-T09	Repeated same image	Stable result under same model/version	Consistency test
FI-T10	Model service unavailable	Application handles failure gracefully	Resilience test
FI-T11	Marketplace hand-off	Structured outputs available downstream	Integration test
FI-T12	Invalid request	HTTP/application error handled	API test

5.5 Deployment and operation

To handle model dependencies without binding them to the Node.js runtime, the ML service is executed independently of the main web application. Starting the ML service, loading the versioned model artifacts, exposing the inference endpoint, configuring the backend integration, and executing the seller workflow constitute the operational procedure. Authenticated service communication, input-size restrictions, logging, monitoring, and controlled model-version promotion should also be used in production deployment.

6. RESULTS AND VALIDATION

6.1 Research results.

The 16-class EfficientNet-B3 fabric classifier has 89.68% test accuracy according to the most recent Fabric Intelligence experiment. Prior to early termination, the best validation accuracy throughout training was around 91.41%. The accuracy of the defect-analysis process is claimed to be 90.87%. These findings show that the models that have been put into practice are capable of learning valuable visual patterns from the given development data.

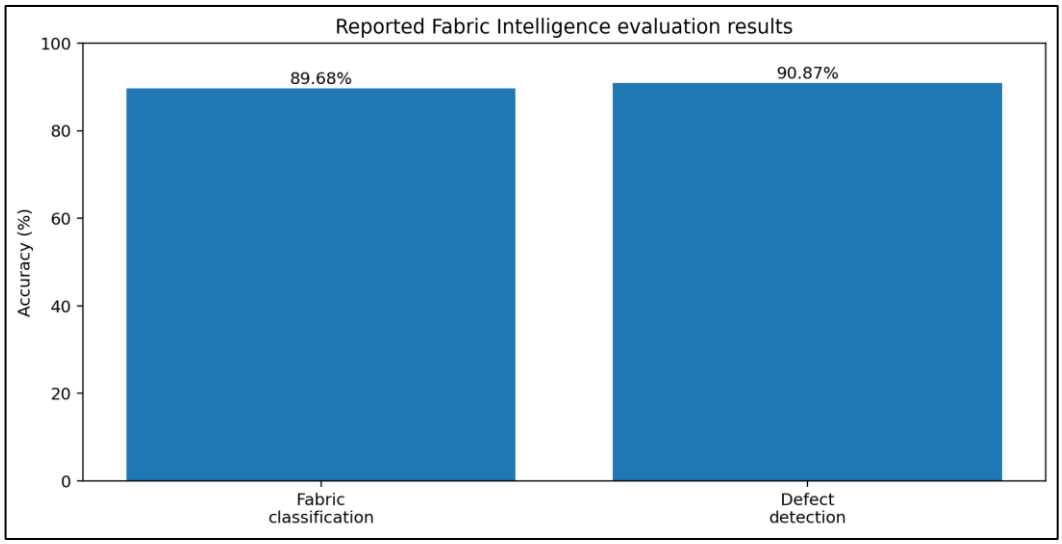


Figure 6.1 Reported Fabric Intelligence evaluation results.

Table 9 : Reported evaluation results

Model	**Evaluation metric**	**Reported result**	**Interpretation**
EfficientNet-B3 fabric classification	Test accuracy	89.68%	Overall accuracy on the documented evaluation set
EfficientNet-B3 fabric classification	Best validation accuracy	≈91.41%	Best recorded validation point before early termination
Defect-analysis pathway	Accuracy	90.87%	Reported binary defect result

6.2 Product and technical results

The model scores are not the only technical outcome. Additionally, the component shows an end-to-end pipeline that includes structured analysis, downstream integration, and picture upload. A class mapping and trained classifier checkpoint are among the project artifacts, and the inference service functions as a software barrier between the models and the application. Instead of leaving the model as a standalone notebook experiment, this makes the component reusable by the whole workflow.

6.3 User or stakeholder validation

Rather than a finished controlled user research, the project evidence that is currently available mainly supports technical and functional validation. Seller-workflow testing verifies that a user may upload a picture, get an analysis, and proceed to further phases of the marketplace. Rather than being created in this thesis, a comprehensive usability study with a statistically supported participant sample is designated as future validation and would be a distinct assessment.

6.4 Integration results

The integration architecture shows that Sustainability Intelligence can use the results of Fabric Intelligence. Specifically, the sustainability component's inputs include fabric type, health score, and reparability. An image-level observation becomes structured data for environmental decision support as a result of this dependency chain. Additionally, the design allows the same standardized textile properties to be consumed by Marketplace Predictive Analytics and Commercial Intelligence.

6.5 Summary of findings

Three conclusions are supported by the facts at hand. First, using the documented assessment data, the 16-class fabric classifier obtains a useful preliminary classification level. Second, a complementary condition signal is sent via the faulty route. Third, a reusable service/application process has been included with the models. However, as actual textile-waste photos may differ in lighting, class balance, fabric mixes, damage patterns, and backdrop circumstances, the results should not be extrapolated beyond the evaluation data without external validation.

7. DISCUSSION

7.1 Interpretation of findings

The classifier learnt significant class-specific visual cues from the assessment data, as evidenced by its accuracy of 89.68%. However, multiclass accuracy by itself cannot identify confused classes. Blended materials can defy the presumption that one picture belongs to one dominating class, and fabrics with comparable textures or weave patterns might be visually challenging to differentiate. Thus, class-level accuracy, recall, F1-score, and confusion matrices should be reported in a more robust evaluation in the future.

Although it does not demonstrate resilience to all defect types, the 90.87% defect accuracy similarly shows meaningful binary discrimination on the stated assessment data. Minor flaws, poor contrast, wrinkles, and lighting abnormalities can affect model confidence. Therefore, rather than automating every forecast, a production system should include a confidence threshold and a manual-review approach.

7.2 Comparison with literature baselines and existing solutions

The chosen architectures align with accepted computer vision practices. While MobileNetV2 was created for effective inference under resource constraints [8], EfficientNet offers an accuracy-efficiency-oriented scaling method [7]. The significance of automated inspection is demonstrated by fabric-defect literature. [4, 5] while textile-specific research like FabricNet shows that image-based CNN techniques can identify textile threads from surface imagery [3]. Therefore, rather than being the creation of a new CNN architecture, the innovation of the current component is better defined as application-level integration and localization inside a circular marketplace workflow.

7.3 Research contribution.

- A project-specific 16-class fabric-intelligence vocabulary integrated into a textile-waste marketplace.
- A transfer-learning pipeline that adapts EfficientNet-B3 to the project classification task.

- A complementary defect-analysis pathway that provides condition evidence.
- A structured transformation from visual predictions to health, repairability, and remaining-life indicators.
- A reproducible model-serving boundary that allows the image models to be consumed by downstream AI components.

7.4 Product contribution

Fabric Intelligence lessens the quantity of manual textile data that a seller needs to supply from a product standpoint. Additionally, it creates a standard machine-readable description of a textile prior to the execution of commercial and sustainability modules. Because downstream components may work on standardized properties instead of separately interpreting the same raw picture, this enhances workflow continuity.

7.5 Limitations and threats to validity

- The assessment's findings should not be used as a standard for all Sri Lankan textile waste since they are based on project development and evaluation data that has been published.
- Not all textile fibers, blends, or treatments are covered by the 16 classifications.
- Lighting, backdrop, folds, occlusion, and camera distance can all have an impact on single-image inference.
- Without the deployment of a proven segmentation component, binary defect detection does not offer full defect localization.
- Rather than being standardized laboratory measures, health, reparability, and longevity are indications obtained from applications.
- A completed user-study result is not established by the available data.
- Before making claims about commercial quality, external confirmation using industry photos is necessary.

7.6 Implications

The immediate conclusion is that expert inspection cannot be replaced by Fabric Intelligence; rather, it may be used as an initial screening layer. Confidence levels, human assessment for ambiguous circumstances, and explicit information that forecasts are advisory should all be included when deploying the component. This location allows for future model development while making the system safer and more believable.

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The TexCycle AI textile-waste marketplace's Fabric Intelligence component was created by Member 01. The component converts image-level evidence into structured textile properties by combining an EfficientNet-B3 multiclass classifier with a different defect-analysis route. 89.68% test accuracy for fabric categorization and 90.87% accuracy for flaw identification are reported in the assessment evidence that is currently accessible. The project shows that computer vision may be used as the initial intelligence layer in a circular textile economy.

8.2 Achievement of objectives

Table 10 : Objective achievement

Objective	**Achievement**	**Evidence**
16-class fabric classification	Achieved	EfficientNet-B3 model and reported 89.68% test accuracy
Defect analysis	Achieved at prototype level	Reported 90.87% binary accuracy
Health/repairability/lifespan outputs	Achieved in application logic	Documented project workflow
ML service integration	Achieved	Python/FastAPI architecture
Seller workflow integration	Achieved	End-to-end workflow documentation
Publication-grade external validation	Not yet achieved	Requires real-world dataset and independent evaluation

8.3 Individual research and product contributions

The research and engineering needed to go from textile-image analysis to an integrated marketplace service are covered by the individual contribution. A trained classification model, defect-analysis capabilities, preprocessing and inference logic, model artifacts, service integration, and evidence-based limitation interpretation are all contributions of this study. Additionally, the component builds the visual-to-structured bridge that other researchers need.

8.4 Future directions

Add more mixes and regional textile materials to the class's lexicon.

- Compile an image dataset of Sri Lankan textile waste from producers, recyclers, retailers, and recovery centers.
- Incorporate multi-image analysis to enable the combination of front, rear, and close-up perspectives.
- Implement clear contamination detection for metal, plastic, stains, and other non-textile materials.
- If accurate defect localization is needed, use verified pixel-level defect segmentation.
- Include a human-review queue and confidence levels.
- Assess resilience under various lighting, backdrop, camera, and textile circumstances.
- For model evaluation, look into explainability techniques like Grad-CAM or saliency.
- Maintain model-version information with marketplace records and version datasets and model artifacts.
- Before claiming commercial performance, carry out an impartial external assessment.

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GLOSSARY

Table 11: Glossary terms

Term	Definition
Accuracy	Proportion of all correct predictions.
CNN	Convolutional Neural Network used for visual pattern learning.
Confidence	Model probability associated with a predicted class in the implemented inference response.
Defect	A condition signal indicating that the defect-analysis model predicts the presence of a defect.
EfficientNet-B3	A member of the EfficientNet CNN family used as the fabric-classification backbone.
Fabric Intelligence	Member 01 component that converts textile images into structured fabric and condition information.
FastAPI	Python framework used to expose machine-learning inference as an API.
Health score	Application-level condition indicator derived from the available visual evidence and project rules.
MobileNetV2	Efficient CNN architecture used for the documented defect-analysis pathway.
Repairability	Application-level estimate of whether the textile is suitable for repair-oriented recovery.
Remaining lifespan	Estimated remaining usable life derived from project-specific baseline and health information.
Transfer learning	Reusing pretrained visual representations and adapting them to a target classification task.

APPENDIX A — FABRIC CLASS SPECIFICATION

Table 12 : Fabric class specification

Index	Class
1	Blended
2	Chenille
3	Corduroy
4	Cotton
5	Crepe
6	Denim
7	Fleece
8	Leather
9	Linen
10	Nylon
11	Polyester
12	Satin
13	Silk
14	Terrycloth
15	Viscose
16	Wool

The project-specific operational mapping recorded for Member 01 is the class vocabulary mentioned above. It is not a comprehensive taxonomy of textile materials or fibers.

APPENDIX B — MODEL SPECIFICATION

Table 13: Fabric Intelligence model specification

Parameter	Fabric classifier	Defect pathway
Task	16-class multiclass classification	Binary defect classification
Backbone	EfficientNet-B3	MobileNetV2 pathway

Input	224×224 RGB	Image preprocessing according to the deployed model
Output	Class + confidence	Defect / No Defect
Training	Transfer learning	Documented lightweight CNN approach
Selection	Best validation checkpoint	Best available documented model
Primary metric	Accuracy	Accuracy

Before publishing, the deployed service must be compared to the faulty implementation boundary. Although the subsequent application description clearly says that the backend does not carry out genuine pixel-level defect segmentation, the research material includes previous references to a segmentation experiment and a repository asset entitled defect_unet.pth. The precise model and endpoint that are implemented should be used in the final proof that is presented.

APPENDIX C — EXTENDED TEST CATALOGUE

Table 14 : Extended test catalogue

ID	Scenario	Expected	Acceptance criterion
FI-T01	Upload supported JPEG	Valid image accepted	Pass criteria: request reaches model service
FI-T02	Upload supported PNG	Valid image accepted	Pass criteria: request reaches model service
FI-T03	Upload unsupported extension	Validation error	No model inference
FI-T04	Empty upload	Validation error	No server crash
FI-T05	Very small image	Controlled preprocessing	No unhandled exception
FI-T06	Large image	Resize/limit handled	No memory failure
FI-T07	Fabric prediction	Class returned	Class belongs to 16-class map

FI-T08	Confidence	Confidence returned	Numeric value present
FI-T09	Defect prediction	Defect/No Defect	Valid binary output
FI-T10	Health score	Health generated	Value within defined application range
FI-T11	Repairability	Repairability generated	Value present
FI-T12	Remaining life	Estimate generated	Value present
FI-T13	Repeat request	Stable output	Same model/version gives stable result
FI-T14	Service unavailable	Graceful application error	No broken UI state
FI-T15	Downstream sustainability	Attributes passed	Member 02 receives required fields
FI-T16	Model version	Version identifiable	Historical result traceable

APPENDIX D — MODEL EVALUATION INTERPRETATION GUIDE

When the final dataset is ready, accuracy and class-level metrics should be supplied. Poor performance in minority classes might be hidden by a single total percentage. A confusion matrix, per-class precision, recall, and F1-score, macro-averaged and weighted-average metrics, and an explanation of the test split should all be included in the final evaluation for fabric classification. When it comes to defect analysis, it is important to differentiate between false positives and false negatives since it might be more serious to overlook an apparent flaw than to ask for a manual inspection of an image that is not flawed.

The established project outcomes of 89.68% fabric accuracy and 90.87% defect accuracy are maintained. Since the raw test predictions were not included in the source information used to create this report, they are not recalculated here.

The established project outcomes of 89.68% fabric accuracy and 90.87% defect accuracy are maintained. Since the raw test predictions were not included in the source information used to create this report, they are not recalculated here.

APPENDIX E — API AND DATA CONTRACT

Table 15 : API and data contract

Field	Type	Purpose
image	File/binary	Uploaded textile image
fabric_type	String	Predicted project fabric class
fabric_confidence	Float	Confidence associated with class prediction
defect_status	String	Defect / No Defect
health_score	Numeric	Application-level condition indicator
repairability	Numeric/categorical	Repairability interpretation
remaining_lifespan	Numeric	Estimated remaining usable life
model_version	String	Version of the model artifact used

Before submitting, the finished code and the deployed property names must be verified. This appendix does not create undocumented endpoint fields; instead, it explains the logical contract that the project documentation supports.

APPENDIX F — CONTRIBUTION TIMELINE

Table 16 : Contribution timeline

Period	Individual contribution
01/04–07/04	Discussed Fabric Intelligence requirements and image-analysis objectives.
08/04–14/04	Investigated textile image datasets and relevant research.
15/04–21/04	Reviewed CNN and transfer-learning approaches for fabric recognition.
22/04–28/04	Investigated preprocessing, class balancing, and dataset preparation.
29/04–05/05	Organized image classes and prepared training workflow.
06/05–12/05	Implemented/experimented with the fabric classification model.
13/05–19/05	Evaluated validation behaviour and model configuration.
20/05–26/05	Reviewed defect-analysis approaches.

27/05–02/06	Integrated the defect pathway and condition interpretation.
03/06–09/06	Prepared model artifacts and class mappings.
10/06–16/06	Worked on ML API integration.
17/06–23/06	Tested image prediction requests and fixed integration issues.
24/06–30/06	Integrated Fabric Intelligence with seller upload workflow.
01/07–07/07	Tested frontend/backend communication and analysis display.
08/07–14/07	Tested different textile images and prediction consistency.
15/07–21/07	Reviewed classification and defect results.
22/07–28/07	Investigated incorrect predictions and improvement opportunities.
29/07–04/08	Gathered Sri Lankan textile/apparel context for research writing.
05/08–11/08	Documented dataset methodology and model training.
12/08–18/08	Prepared evaluation evidence and presentation material.
19/08–22/08	Finalized current model evaluation, viva explanation, and evidence organization.

APPENDIX G — SUGGESTED EVIDENCE CAPTURE PLAN

- Before analysis, record the seller image-upload screen.
- Record the Fabric Intelligence outcome that displays the kind and confidence of the fabric.
- Record the outcome of the defect or disease and the resulting health data.
- When reparability and remaining-life outputs are shown, record them.
- If allowed, record the network/API response or service log demonstrating successful inference.
- Once Member 01 outputs are transmitted downstream, capture the Sustainability Intelligence screen.
- Only capture the model/service configuration when it doesn't reveal private passwords or secrets.
- Make use of figure captions that describe the screenshot's function, setting, and feature.

APPENDIX H — REPRODUCIBILITY CHECKLIST

Table 17 : Reproducibility checklist

Item	Required evidence
Dataset	Dataset source and class mapping documented.
Class mapping	fabric_classes.json or equivalent
Preprocessing	Image size, RGB conversion, and normalization documented.
Training	Optimizer, schedule, and stopping criteria documented.
Checkpoint	Selected model artifact retained
Evaluation	Test split and metrics retained.
API	Endpoint contract verified against code.
Frontend	Seller workflow tested
Integration	Downstream hand-off tested
Versioning	Model version associated with results
Evidence	Actual screenshots inserted
References	Every in-text citation appears in reference list.

APPENDIX I — RESPONSIBLE AI AND DEPLOYMENT CHECKLIST

- Don't use laboratory-confirmed fiber composition to represent the anticipated fabric class.
- As a controlled industrial quality rating, they do not reflect health or reparability.
- Manually review cases that are unclear or low-confidence.
- Only save the picture data required for the specified process.
- Stop illegal access to textile photos that have been posted.
- For auditability, record the model version and inference time.

- If there are significant changes to the dataset or class distribution, reevaluate the model.
- Before making any business claims, verify against actual industry photos.

APPENDIX J — RESEARCH INTEGRITY AND AI-USE DISCLOSURE

Software development and document preparation technologies were used in the creation of this thesis. Drafting, organizing, language refining, and formatting may have all benefited from generative-AI support. The candidate is still in charge of evaluating citations, confirming stated metrics, comparing any technical claims to the implemented system and source materials, and making sure that no false proof is offered. Only screenshots from the tested system should be used.

A claim has been qualified as a prototype result, a documented design choice, or a future-validation need when it could not be proven using the project evidence that was available. For the contrast between development-set precision and actual industrial performance, this is very crucial.

APPENDIX K — DATASET QUALITY AND PREPARATION PROTOCOL

K.1 Dataset objectives

Binary defect analysis and multiclass fabric recognition are two distinct visual-learning tasks that must be supported by the Fabric Intelligence dataset. Because each task requires different labels, class balance, and visual evidence, the preparation procedure should maintain the separation between these tasks. While a defect classifier requires enough positive and negative samples spanning actual defect appearances, a fabric classifier requires representational variance within each material class.

K.2 Image inclusion criteria

- The main visual topic of the image must be a textile or an item related to textiles.
- The image must be legible enough to recognize building elements or textures.
- Rather than being quietly included, images with significant corruption should be eliminated or marked individually.
- There shouldn't be any near-duplicate or duplicate photos in the training and test partitions.
- When accessible, images should maintain their original class identification and dataset provenance.
- Training data and images needed for external validation must be kept separate.

K.3 Class-balance protocol

A model may favor visually similar groups due to class imbalance. Class filtering and balancing are thus part of the project procedure that is described. The train/validation/test split, the number of pictures per class, and any augmentation approach should all be clearly disclosed for the final research release. If oversampling is used for class balancing, the duplication must be limited to the training partition to maintain assessment independence.

Table 18 : Dataset quality checks

Quality check	**Purpose**	**Evidence to retain**
Duplicate check	Prevent train/test leakage	Duplicate report or hash list
Class count	Assess imbalance	Count per class
Resolution check	Detect unusable images	Resolution summary
Colour mode	Ensure consistent input	RGB conversion rule
Label review	Reduce annotation noise	Manual review log
Split integrity	Protect evaluation validity	Train/validation/test manifest

K.4 Dataset shift risks

Changes in lighting, backdrop, camera quality, textile folding, distance, occlusion, and material combinations are the most significant dataset-shift concerns. Stains, labels, seams, printing, and many textiles in one frame may also be found in industrial photos. A classifier that works well on a controlled development dataset may become less reliable due to these variables. Therefore, rather than focusing just on aesthetically pleasing instances, external evaluation should purposefully include challenging scenarios.

APPENDIX L — IMAGE PREPROCESSING AND INFERENCE PROTOCOL

L.1 Preprocessing sequence

The basic preprocessing presumptions employed during training must be replicated in the inference pipeline. 224x224 RGB input is used by the described fabric classifier. Therefore, a robust implementation loads the picture, verifies its format, converts it to RGB, resizes it to the resolution of the model input, performs the required normalization, transforms it into a tensor, and adds the batch dimension.

Table 19: Image preprocessing sequence

Step	Operation	Failure risk	Control
1	Load image	Corrupted file	Controlled exception
2	Convert RGB	Unexpected colour mode	Explicit conversion
3	Resize 224×224	Distortion	Consistent resize policy
4	Normalize	Train/inference mismatch	Shared preprocessing
5	Tensor conversion	Wrong shape	Shape assertion
6	Batch dimension	Runtime shape error	Batch validation
7	Model inference	Missing model/device	Startup validation
8	Decode class	Wrong mapping	Versioned class map

L.2 Confidence handling

The major prediction is the class with the highest likelihood. However, unless calibration has been specifically assessed, confidence should not be understood as a calibrated likelihood of real-world accuracy. Determining a confidence threshold from validation tests and directing ambiguous circumstances to manual review is the suggested operational strategy.

L.3 Robustness checks

- Examine the identical image with various file encodings.
- Examine appropriate adjustments to the image's resolution.
- Examine photos with a moderate range of brightness.
- Examine fabrics that have been partially folded.
- Examine variations in the background.
- Examine visually similar classes.
- Examine mixed textiles and clearly note whether a single dominating class is anticipated as the proper result.

APPENDIX M — TRAINING AND EXPERIMENT RECORD

M.1 Training design

Transfer learning using an EfficientNet-B3 backbone is used in the stated training method. The final classification layer is tailored to the project's 16 classes, while the backbone offers a pertinent visual representation. Early halting, learning-rate scheduling, and AdamW optimization are all part of the training. The checkpoint that has the best validation performance is the one that was chosen.

Table 20 : Training design

Experiment element	**Documented approach**	**Reason**
Backbone	EfficientNet-B3	Strong transfer-learning candidate
Input	224×224 RGB	Consistent deployment contract

Optimization	AdamW	Documented training choice
Scheduling	Learning-rate scheduling	Improve convergence control
Stopping	Early stopping	Limit overfitting/unnecessary training
Selection	Best validation checkpoint	Deployment candidate
Evaluation	Test accuracy + class metrics recommended	Separate generalization evidence

M.2 Experiment log template

Table 21 : Experiment log template

Run	**Change**	**Validation accuracy**	**Test accuracy**	**Decision**
Run 01	Baseline transfer learning	Record	Record	Baseline
Run 02	Class balancing	Record	Record	Compare
Run 03	Augmentation adjustment	Record	Record	Compare
Run 04	Learning-rate adjustment	Record	Record	Compare
Run 05	Final selected checkpoint	≈91.41% documented	89.68% documented	Selected

M.3 Interpretation

The given values show that, under the described assessment environment, the chosen model generalizes rather well. It is normal for there to be a discrepancy between test and best validation accuracy; this discrepancy should not be interpreted as overfitting. The size and makeup of each split, class-level findings, multiple runs, and, if feasible, confidence intervals are all necessary for accurate interpretation.

APPENDIX N — ERROR TAXONOMY AND CONFUSION-MATRIX PROTOCOL

N.1 Fabric classification errors

Instead of only being regarded as inaccurate labels, fabric categorization failures should be organized into relevant visual reasons. Texture similarity, blended materials, low picture quality, folds, occlusion, background interference, unconventional presentation, and annotation ambiguity are all useful categories.

Table 22 : Fabric classification error taxonomy

Error category	Example	Recommended response
Texture similarity	Corduroy vs another ribbed texture	More representative training samples
Blend ambiguity	Blended material predicted as dominant pure class	Add blend classes / multi-label research
Occlusion	Only part of textile visible	Multi-image or better framing guidance
Lighting	Colour/texture altered by illumination	Augmentation and robustness data
Background	Complex background dominates image	Segmentation/cropping or seller guidance
Annotation	Image label inconsistent with visual evidence	Label review

N.2 Confusion matrix

Once raw test predictions are provided, the final experimental evidence should contain a confusion matrix. Actual classes should be represented by rows, and anticipated classes should be represented by columns. When necessary, totals and normalized percentages should be included. The matrix is particularly helpful in figuring out whether a few dominating classes are responsible for the reported overall accuracy.

N.3 Defect errors

False negatives are crucial for defect analysis because they might spread an unduly optimistic condition interpretation if a flaw is overlooked. The fact that false positives might direct useful resources toward needless manual examination makes them significant as well. Therefore, the operational cost of both types of errors should be the basis for threshold tuning.

APPENDIX O — API SEQUENCE AND FAILURE MODES

O.1 Logical request sequence

Table 23: Logical API request sequence

Step	Actor	Action	Expected response
1	Seller UI	Select image	Image available locally
2	Frontend	Submit image	HTTP request created
3	ML service	Validate file	Valid input or controlled error
4	ML service	Preprocess image	Tensor created
5	ML service	Run fabric model	Class probabilities
6	ML service	Run defect model	Defect result
7	ML service	Derive condition fields	Structured result
8	Backend	Persist/use result	Application workflow continues
9	Sustainability module	Consume relevant attributes	Sustainability analysis

O.2 Failure-mode matrix

Table 24 : API failure-mode matrix

Failure	Likely cause	User impact	Mitigation
Upload rejected	Unsupported format/size	Cannot analyze	Validation message
Model unavailable	Service stopped	Analysis unavailable	Health check + graceful UI

Model artifact missing	Deployment error	Inference fails	Startup validation
Class map mismatch	Version mismatch	Wrong label	Versioned artifacts
Timeout	High image/model load	Delayed result	Timeout + retry policy
Malformed response	API contract mismatch	UI error	Schema validation
Low confidence	Ambiguous image	Uncertain class	Manual review threshold

O.3 Operational logging

Request time, model version, processing result, and error type should all be recorded in logs without needlessly retaining critical picture material. Because further retraining may alter predictions for the same picture, model-version recording is crucial. Therefore, a marketplace record needs to be able to identify the model that generated its analysis.

APPENDIX P — SECURITY AND PRIVACY CHECKLIST

The idea that uploaded textile photos can include commercially sensitive information should guide the design of the image-analysis methodology. Applications, APIs, and storage layers should all have security controls.

- Before permitting the retrieval of confidential analytical records, authenticate the user.
- Permit access to prevent one seller from accessing the private textile photos of another vendor.
- Verify file types and impose upload size limits.
- Don't run submitted material as code.
- When deploying APIs, use secure transport for communication.
- Avoid needlessly logging raw picture data.
- Keep model artifacts out of places that are accessible to the public.
- The source code should not contain service secrets.

- Keep track of model iterations without disclosing internal infrastructure information to regular users.
- Establish a retention policy for analytical records and submitted photos.

P.1 Responsible AI boundary

Decision support is the component. It is not appropriate to portray a model prediction as a professional quality grade, safety certification, or guaranteed material composition. For instances that are unclear, valuable, or contentious, human review should still be an option.

APPENDIX Q — FABRIC INTELLIGENCE MODEL CARD

Table 25 : Fabric Intelligence model card

Field	Description
Model name	TexCycle Fabric Classifier — EfficientNet-B3
Task	16-class image classification
Input	224×224 RGB textile image
Output	Fabric class and confidence
Development metric	89.68% documented test accuracy
Best validation metric	Approximately 91.41% documented.
Known risks	Class ambiguity, blends, lighting, background, folds, occlusion
Intended use	Preliminary marketplace textile characterization
Not intended for	Laboratory fiber certification or regulated quality inspection
Human oversight	Recommended for low-confidence and unusual cases
Update policy	Re-evaluate after major dataset or class changes.
Versioning	Model artifact and class mapping should be versioned together.

Q.1 Defect model card

Table 26 : Defect model card

Field	Description
Task	Binary defect / no-defect classification
Architecture	Documented MobileNetV2 pathway
Reported result	90.87% accuracy
Intended use	Preliminary condition evidence
Limitation	Not equivalent to pixel-level defect segmentation
Human oversight	Recommended for uncertain cases
Future	Validated localization/segmentation if required

APPENDIX R — OBJECTIVE-TO-EVIDENCE MAPPING

Table 27 : Objective-to-evidence mapping.

Research objective	Evidence in thesis	Further evidence needed
Develop fabric classifier	Ch. 3, Ch. 5, Appendix B/M	Training logs and checkpoint metadata
Evaluate classification	Ch. 6	Per-class precision/recall/F1 and confusion matrix
Develop defect analysis	Ch. 4, Ch. 5, Ch. 6	Raw test predictions and class-level metrics
Generate condition indicators	Ch. 4	Actual application screenshots
Integrate ML service	Ch. 4, Ch. 5	API logs/endpoint screenshots
Integrate seller workflow	Ch. 5, Ch. 6	Actual UI evidence
Support sustainability module	Ch. 5, Ch. 6	Cross-component integration evidence
Document limitations	Ch. 7, Appendix I/Q	External validation study

R.1 Traceability from input to decision

The minimum chain is image identifier → preprocessing/model version → predicted class → defect result → health/reparability interpretation → downstream analysis. This chain is useful for debugging, auditability, and future research because a later model can be compared against the exact historical output. A seller image should remain traceable through the stages of analysis.